



Freddie Mac Loans Analysis

Introduction

Nationwide, homeownership remains a cornerstone of the American dream, and mortgage lending plays a central role in making that dream accessible. Every year, millions of Americans apply for mortgage loans to purchase homes, remodel existing properties, or fund improvements on their businesses. Behind the scenes, large portions of these loans are evaluated, purchased, or guaranteed by government funded corporations like Freddie Mac and Fannie Mae. These corporations aim to increase access to homeownership while ensuring stability in the housing finance system. In order to do that, Freddie Mac and these other corporations collect detailed data on the loans they acquire, including borrower demographics, income, property characteristics, and loan terms.

This analysis uses a dataset derived from Freddie Mac-backed mortgage loans across the United States. The dataset is made up of 377995 rows and 71 columns. Out of the seventy-one columns, eight of them are of the dtype 'float64', four of them are of the dtype 'object', and fifty-nine of them are of the dtype 'int64'. The non-null-counts in each column are all equal to the amount of rows there is in the dataset, meaning that there are no missing values.

The dataset is a 10% stratified sample based on states, ensuring that each state is represented proportionally to the number of Freddie Mac loans in that region. The sample includes different features related to the borrowers, properties, and loan characteristics, including demographic, financial, and geographic information. This 10% stratified sampling approach allows for meaningful information that is reflective of the larger population of Freddie Mac borrowers which is 100% of the original dataset.

The dataset includes information from merged files as well. The presence of multiple columns in the dataset such as MSANAME, MSAPos, and MSANAME1 with the dtype "object" and 'int64' shows that these were merged from the MSA_2020 txt file to assign geographic locations to each mortgage record. This dataset is very valuable for understanding the demographics and financial profiles of Freddie Mac mortgage borrowers. It will allow for a deep dive into state-by-state variations in borrower characteristics, loan types, and financial health. In particular, by analyzing the geographic location and demographic factors, we can uncover patterns in loan accessibility and identify any key differences or anything that stands out across different states in the US.

Data and how we acquired it

I obtained the sample file titled fhlmc_sf2020c_loans_samp10.csv, which includes key borrower, loan, and property-level attributes. To generate the graph displaying the top ten states with the highest number of borrowers, I used a pd.crosstab of the STATE variable to count the number of loans per state. From this count, I isolated the ten states with the highest borrower counts and created a binary indicator feature to mark the states that are in the top 10 with a "1" and the rest of the states with a "0".

A quick look at the sample data reveals that California has by far the largest share of Freddie Mac-backed loan borrowers, representing 15.6% of all borrowers in the dataset. In a part of the following analysis, we will explore possible reasons why California accounts for such a significant portion of borrowers in this sample. CA has more than double the next highest state, Texas, which accounts for 6.4%. Following Texas, Florida ranks third with 5.6%, and Illinois comes in fourth at 4.4%.

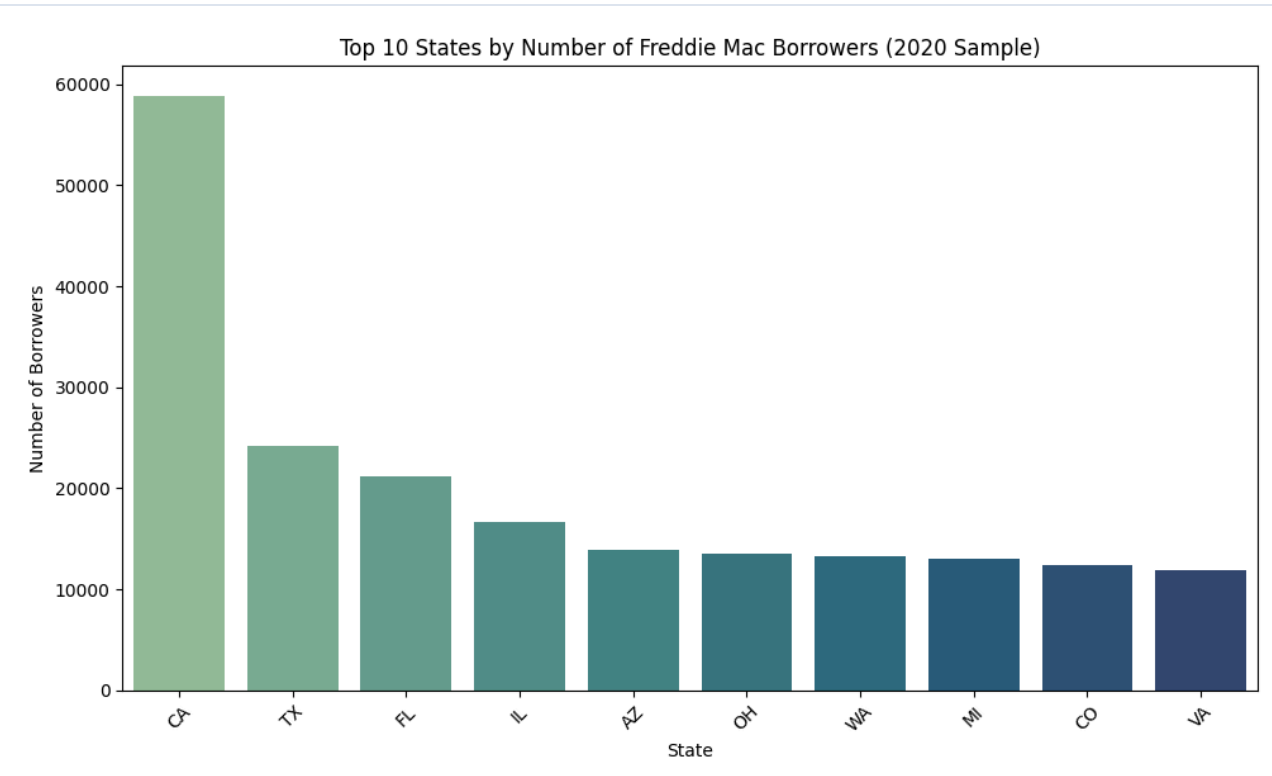
Then I created a graph to display the top ten states in the x axis and the amount of borrowers in the y axis.

Unlike the data in the table, which emphasizes percentages, the bar chart allows for a more intuitive comparison of absolute borrower counts. California's dominance is clearly shown, with nearly 60,000 borrowers—visibly far ahead of the other states. This graph allows puts into perspective how many individuals are being represented by each state in this dataset.

```
/tmp/ipykernel_255/940175748.py:2: FutureWarning:
```

```
Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False`
```

```
sns.barplot(x=top_states.index, y=top_states['Borrower_Count'], palette='crest')
```



Analysis

In order to conduct a deeper dive into the states loan borrowers' demographics I generated 2 mean-based profiles for the states with the highest amount of borrowers. That would be California with 58,856 borrowers and Texas with 24,216 borrowers. In order to conduct these mean based profiles, I selected seven key features from the dataset that would help me understand more about the demographics of the borrowers. I chose the features of the borrower's gender, age, annual income, the size of the loan they received, debt to income ratio, occupancy code, and the tract income ratio.

When comparing the profile for both states relating to the borrower gender feature, California has a mean of 1.5 and Texas of 1.4. When looking at the original dataset I noticed that the gender of the borrowers is represented by 1 and 2. Not specifying which gender is assigned to which number but I can still conclude that with the mean being closed to 1.5 for both states, there is a near-even distribution between male and female borrowers.

When comparing the borrower age feature for both states, California has a higher mean(3.89) than Texas (3.49) possibly meaning that slightly older individuals are qualifying for loans in California than in Texas. The unique values in the Age column are numbers 1-9 (with the exception of 8) and after doing some research and looking at the SF Census Tract Dictionary File, Freddie Mac does not explicitly divide age groups into 1-9 for loan eligibility purposes, therefore that is the only assumption I can make based on the mean.

Next I looked at the Loan Note Amount and Borrower's Income features closely together. Borrowers in California have a higher average annual income (\$143,480) and loan amount(\$404,143) compared to those in Texas. While the average income for borrowers in Texas(\$137,635) is only around 6 thousand less than in California, the average loan amount is about \$140,000 less. This likely indicates how the cost of owning a home in California is much higher than in Texas and that is a key factor in determining loan amounts, because if the loan amount was based on the annual income, they would be much closer to each other since the income is not significantly different.

Next I looked at the feature Tract_Local_Median_Inc_Ratio, which according to the SF Census Tract Dictionary File this is the ratio used to determine whether the census tract qualifies as a low-income area for purposes of the single-family low-income areas housing goal. The mean for each state is above 7 which I found a little weird because the example in the SF Census file says "Example of data format: 1.7500 = The tract income is 175 percent of the local area median income." If we follow this definition then that means the average income would be 700% of the local median which is not likely. Therefore I used the function .describe() to obtain the min,median,1st and 3rd quartile, and the max value in that column. 75% of the values fell under 1.5 meaning that a mean of 7.8 is definitely not correct. I seen that the max was 9999.00 but according to the dictionary, Freddie Mac Loans uses that number as a placeholder for "not able to code" therefore those are outliers making the mean of the dataset inflated. To deal with this issue I created a new dataset "loan_df_clean" which removed any values greater than 10 from the Tract_Loca_Median_Inc_Ratio column, and my new max was 3.9 which confirms that the number inflating the mean was the 9999.00 since the highest number other than that one was 3.9. I then conducted a new mean based profile between the two states and received a mean of 1.2 for California and a mean of 1.3 for Texas. This means that Texas borrowers tend to live in areas with slightly higher income levels relative to local medians.

Next I focused on the Occupancy code feature which shows what the property is being occupied for. The SF Census Tract Dictionary file does not explain what the values 1-3 indicate in this column, therefore after some research I found that according to the FFIEC Home Mortgage Disclosure Act website for all loan mortgage organizations the 1 in the Occupancy Code section indicates Primary Residence, 2 is for Secondary Residence, and 3 is for Investment Property. Both Texas and California have nearly identical means for this feature, California has 1.148 and Texas has 1.145. This suggests that majority of the borrowers in both states obtain the loan for a primary residence.

I then looked at the Debt-to-Income (DTI) Ratio as the final feature in my mean-based profile of the two states. California has a slightly higher average DTI at 32.3 compared to 30.9 in Texas. This could indicate either higher overall debt levels among borrowers in California compared to those in Texas.

Mean-Based Profiles for Two States:		
	CA	TX
BORROWER_GENDER	1.500034	1.433143
BORROWER_AGE	3.898481	3.498596
BORROWER_ANNUAL_INCOME	143480.936523	137635.199868
TRACT_LOCAL_MEDIAN_INC_RATIO	7.858471	7.140385
NOTE_AMT	404143.757289	264093.986951
OCCUPANCY_CDE	1.148311	1.145317
DEBT_TO_INCOME_RATIO	32.357517	30.910307

Mean-Based Profiles for Two States (Cleaned File):		
	CA	TX
BORROWER_GENDER	1.500026	1.433146
BORROWER_AGE	3.898686	3.498802
BORROWER_ANNUAL_INCOME	143445.500451	137620.237997
TRACT_LOCAL_MEDIAN_INC_RATIO	1.233609	1.360448
NOTE_AMT	404160.871500	264055.449467
OCCUPANCY_CDE	1.148171	1.145277
DEBT_TO_INCOME_RATIO	32.356275	30.910751

Results

All in all, to better understand the borrower characteristics behind Freddie Mac–secured loans, I generated mean-based demographic profiles for California and Texas—two of the states with the highest number of borrowers in the dataset. The comparison focused on several borrower features, including income, age, debt-to-income ratio, and property occupancy type. These features made it possible to identify patterns and differences in the data between the two states.

These patterns align with current economic and real estate trends: California's housing market is more expensive and competitive, often requiring larger loans and leading to a higher debt burden. Texas, with a lower cost of living and more affordable housing, shows slightly more conservative borrowing profiles. These findings illustrate how regional factors can impact borrower profiles and lending patterns in the mortgage market.

The bar chart illustrates a comparison between the average borrower annual income and the loan (note) amount for Freddie Mac–backed borrowers in California and Texas.

